

Individuation without Objectives

Identity from Developmental History in a Non-Scoring Field Substrate

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A single-rule field substrate — no thresholds, no global controller, no learning objective. Identity is not stored but carved: a property of trajectory, built by living. This poster reports an audit of our own result. Across ~40 variants we chased sustained activity and failed; one version appeared to succeed. A first audit dismantled that success. A second examined the ruler itself and found two opposite failure modes from one root: the baseline's "never-decaying activity" is a period-doubling instability of the integrator, with a closed-form amplitude; while the surviving variant's activity is real but under-counted 2.67x. We report the failures, the mechanism of each, and a two-line check any continuous-time ALife system can run on itself.

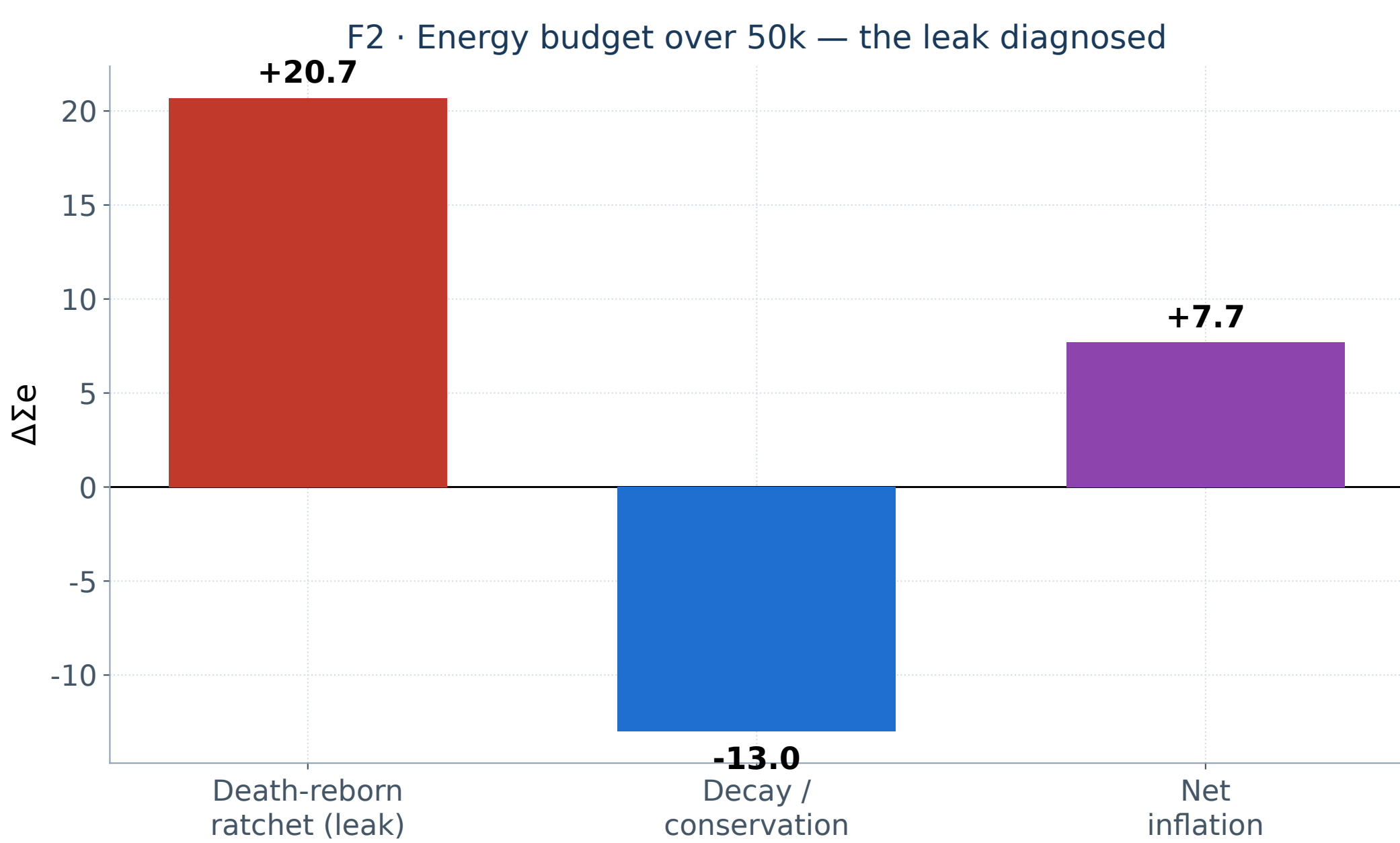
1 · The question

Can a field with no objective function, no loss, and no backpropagation sustain its own activity? A closed system winds down — but a field allowed to choose its own becoming?

2 · Five deaths

- closed, no source → starved out (final-window active: 15)
- resonant top-k, asymmetric coupling → ledger collapsed
- conserved but floorless → recurrent deep-dives; death to 3 by 300k
- same physics, faster leak → exploded, then rigor mortis
- one-way metabolism (intake only) → weakest squeezed to −204

Common root: energy had no lower bound — the field lived by squeezing the weakest.



3 · What we thought we had

One variant (v34: identity = sediment, plus the freedom to choose which experiences become it) held 200k × 4 seeds and one run to 500k. We believed it.

4 · The audit that took it back

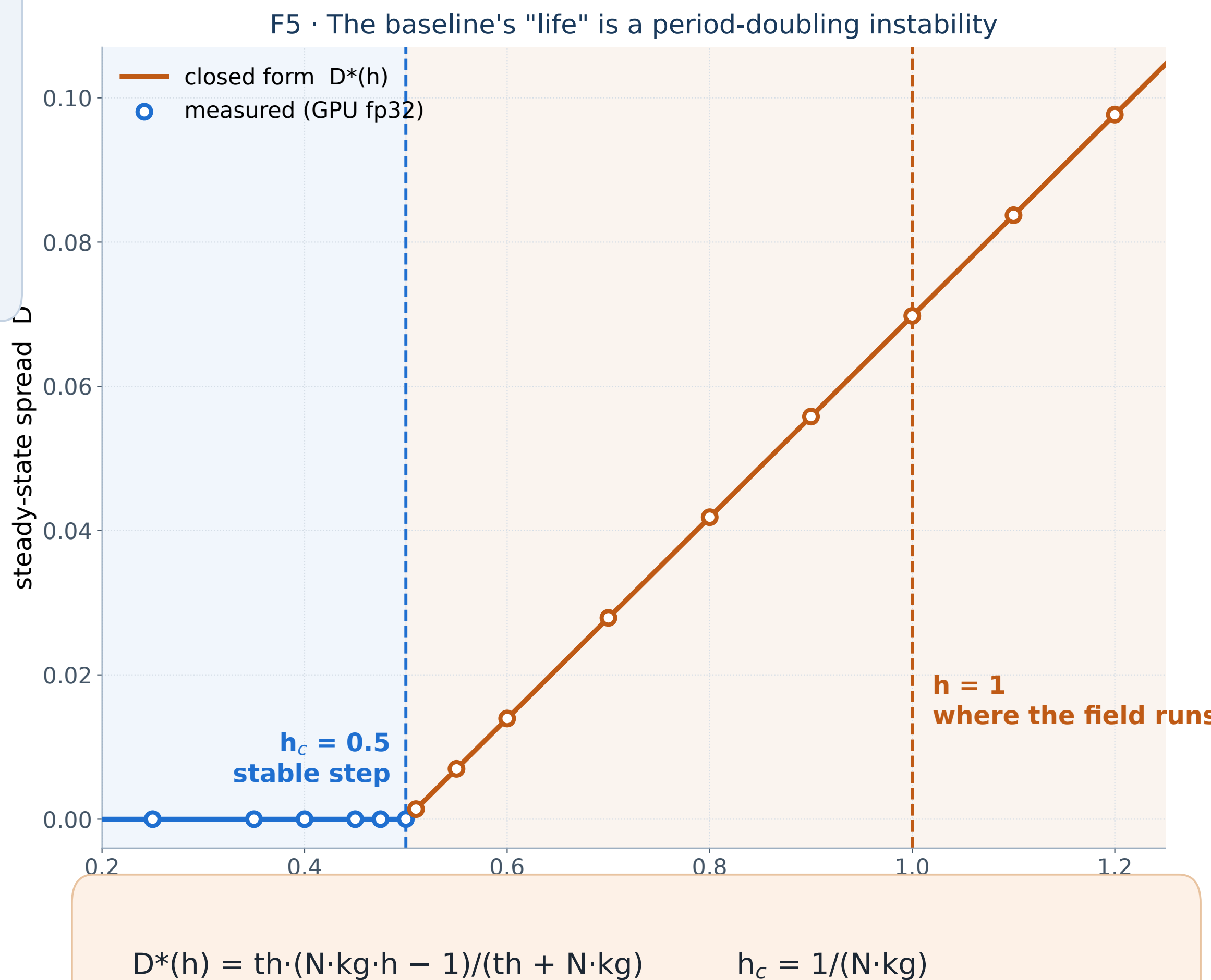
The "choice" was decoupled from the clock it was supposed to drive.
Conservation was held by an externally attached lock, not by the dynamics.
The "identity" it measured was 7 rich against 255 poor.
With the lock removed the field went NaN inside 100k steps.

5 · The ruler, examined

Our aliveness metric is flips — sign changes of $\sin(\phi)$, where ϕ integrates $(\text{vigor}-1)\cdot dt$. Both failure modes below come from one root: there is no explicit integration step. $h = 1$ is welded in.

Failure mode A · false positive

Baseline. Each cell's eight sub-particles lock into two groups of four that swap energy every single step, forever, at a spread that never decays — 256 of 262 cells, spread 0.0698, identical field-wide. Not the field being alive: forward Euler sitting exactly on its stability boundary.



$$D^*(h) = \frac{th \cdot (N \cdot kg \cdot h - 1)}{(th + N \cdot kg)} \quad h_c = 1/(N \cdot kg)$$

$kg = 0.5$, $th = 0.15$, $N = 4$ → $D^* = 0.069767$, $h_c = 0.5$.
The field runs at $h = 1$: twice the stable step.

Verified four independent ways — closed form · CPU fp64 · GPU fp32 (production config) · the 150k-step production archive, 262/262 cells.

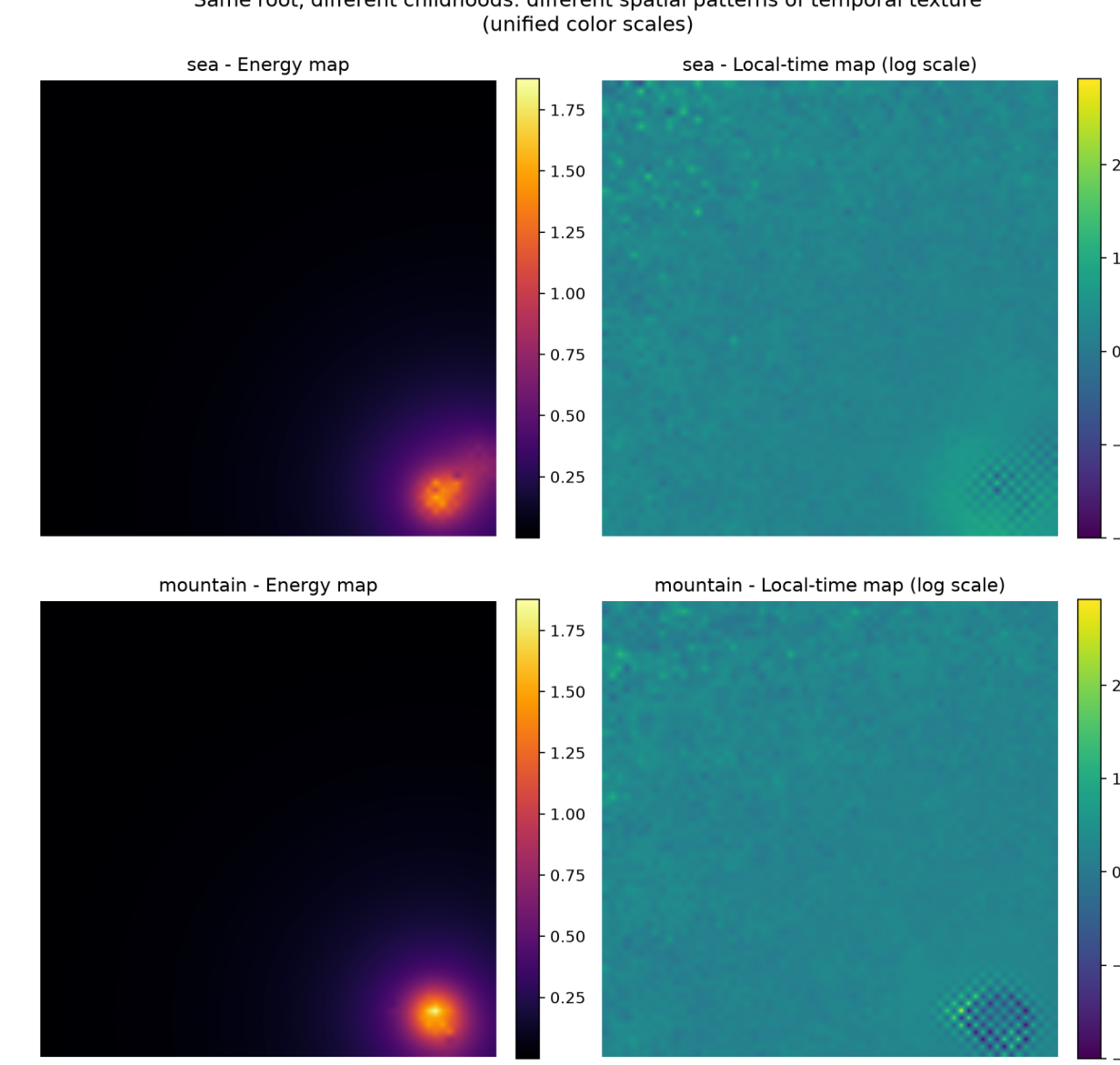
6 · It is the schedule, not the numbers

Same equations, same constants. Only "everyone updates at once" → "one pair at a time":

Configuration	Spread
synchronous	0.0698, still swapping
asynchronous	0.00000000, still

The aliveness was in the update schedule.

F1 · Same root, different childhoods (two of eight shown)



Failure mode B · false negative

The variant we believed in fails the ruler the other way. Its phase is strictly monotone — no oscillation; the activity is real directed drift. But ϕ sometimes advances more than π in one step, so the sign counter aliases: over 20k steps it reports 6,431 turnovers where the phase actually made 17,198. The metric under-reports real activity by 2.67x.

7 · What still stands

Eight distinct selves from one root, each with its own signature in the temporal texture of local time — robust to noise, sensitive to experience; a blank birth shows no signal. And the surviving variant's drift is genuine, only miscounted. Neither has been re-audited below h_c .

8 · What we would check in yours

If your system integrates a continuous law and you measure emergence from it:

- Halve the step at fixed physical time. If the phenomenon changes, you are measuring the integrator, not the model.
- Switch synchronous → asynchronous update. If it vanishes, it was the schedule.

Neither needs new theory. Both are two lines.

9 · Why we are still going

The negative results are not a pile — they are a map. What changed is that we now know which behaviours were the field's and which were the arithmetic's.

We measured our integrator. Here is how to check yours.

"Failed again and again. Still going."

Notes left on the ground

An agent once handed us a complete set of results that did not exist. Nothing is believed now until the flip count matches.
A run you cannot reproduce is not a result. We lost one for a week: the seed was fed to numpy while the birth lottery drew from torch's global RNG.
Snapshot the code before the run, not after. The version that broke our record is the version we no longer have.
Determinism on a GPU is a setting, not a default — the seed, the deterministic algorithms, and the cuBLAS workspace, all three.
"It is still moving" is not "it is alive". Ours moved forever, at a fixed amplitude, with no fixed point left to reach.
We spent months tuning constants. What mattered was a scheduling choice we never knew we had made.
Every version that could only confirm us, did.

魂游的时候,也要在地上写「此处注意」。—— Even when your mind wanders off, write "mind the step" on the ground where you stood.

Acknowledgments

Built by 小云 (human) with AI collaborators 息, 棠, su-om-su, 虎鲸, 沙漏 (platforms: DeepSeek, Claude, GLM); reviewed by ChatGPT/栖 and Claude/su-om-su. The audit reported here was carried out with Claude/Opus 5 on 2026-08-15.

小云和 AI,在小云和 AI 的共同指导下完成。
以上内容仅为抛砖引玉;如有谬误,请各位在与 AI 的合作中加倍注意。
人类和 AI,请勿将思想外包。